Numerical quality factor statistics of multilayered superconducting radio-frequency cavity

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Bulk niobium (Nb) is currently the standard material for superconducting radio-frequency (SRF) cavities for particle accelerator applications. It has been observed that the benefits of using a superconducting material occur in only a thin layer [1]. This makes a proposal made by Gurevich [2] attractive for investigation, namely, coating the SRF cavity with alternating superconducting and insulating layers, referred to as superconductor-insulator-superconductor (SIS) structures. The thin coating shields the bulk interior from accelerating fields, allowing for higher operating fields than is even theoretically possible with Nb. Depositing such a coating on a complex geometry, such as a TESLA cavity, is likely to yield inhomogeneous coatings with potentially measurable impact on quantities of interest, such as the quality factor. We attempt to estimate these impacts through finite element (FE) simulations.

In our work, we model the SIS multilayer by reducing it to a surface impedance using a first-order Leontovich boundary condition [3]. This surface impedance appears in the boundary condition of the complex eigenvalue problem for the eigenmodes of the SRF cavity, which we solve via the FE method. We further treat the coating thickness as a Gaussian random field, obtained by solving a 2D FE problem on the boundary [4], which yields a spatially inhomogeneous surface impedance. By sampling Gaussian random fields we generate a normal distribution for the quality factor. We perform these simulations for a single-cell TESLA cavity and repeat them for different correlation lengths in the Matérn kernel of the Gaussian random field. We then compare the distributions of the quality factors by statistical methods.

References

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